

AI AGENTS

Development Guide — Strategies, Architectures & Best Practices

A comprehensive reference covering everything you need to design, build, and deploy AI agents — from foundational concepts to advanced multi-agent orchestration, with honest analysis of pros, cons, and trade-offs for every major strategy.

Topic	Coverage
Foundations	What is an AI agent, components, types, mental models
Reasoning Strategies	ReAct, Chain-of-Thought, Tree-of-Thought, Self-Reflection
Agent Architectures	Single-agent, Multi-agent, Hierarchical, Swarm, Pipeline
Memory Systems	In-context, External Vector DB, Episodic, Semantic, Procedural
Tool Use & Function Calling	Tool design, function calling, API integration, sandboxing
Orchestration Frameworks	LangChain, LangGraph, AutoGen, CrewAI, Semantic Kernel, Llama Index
Planning Strategies	Zero-shot, Few-shot, Task decomposition, MCTS, self-play
RAG for Agents	Naive RAG, Advanced RAG, Agentic RAG, GraphRAG
Safety & Reliability	Guardrails, hallucination control, prompt injection, testing
Production Deployment	Observability, cost control, latency, scaling, evaluation

1. FOUNDATIONS — WHAT IS AN AI AGENT?

An AI Agent is an autonomous system that perceives its environment, reasons about goals, takes actions using tools, observes outcomes, and iterates — all without requiring human intervention at each step. Unlike a simple LLM API call (prompt → response), an agent operates in a loop: Perceive → Think → Act → Observe → Repeat.

1.1 The Agent Loop

Every AI agent — regardless of framework or complexity — follows some variation of this loop:

- Perceive: Receive input (user query, tool result, environment observation, sensor data)
- Think / Plan: Use LLM or reasoning engine to decide next action (or final answer)
- Act: Execute action — call a tool, query a database, run code, send a message
- Observe: Receive result of action and add to context
- Loop: Repeat until goal achieved or stopping condition met

1.2 Core Components of Every Agent

Component	Purpose	Examples
LLM Brain	Reasoning, planning, generation	GPT-4o, Claude 3.5, Gemini, Llama 3
Memory	Store context, facts, past actions	In-context window, Vector DB, Redis
Tools	Interact with external world	Web search, code exec, API calls, DB queries
Planner	Decompose goals into sub-tasks	ReAct, CoT, ToT, Task graph
Executor	Run actions safely	Function calling, sandboxed code, subprocess
Evaluator	Assess output quality / progress	Self-critique, reward model, human feedback
Orchestrator	Coordinate multi-agent systems	LangGraph, AutoGen, CrewAI

1.3 Agent vs. Chatbot vs. Workflow

Dimension	Chatbot	Workflow/Pipeline	AI Agent
Autonomy	Low — responds only	Medium — fixed steps	High — self-directed
Loop	Single turn	Fixed sequence	Dynamic, adaptive
Tool use	Minimal/none	Pre-defined integrations	Dynamic tool selection
Goal handling	Reactive	Pre-specified	Goal-directed
Memory	Conversation history	Shared state	Multi-layer memory

2. REASONING STRATEGIES

How the agent thinks is the most critical design decision. Different strategies suit different problem types, cost envelopes, and latency requirements.

2.1 ReAct (Reasoning + Acting)

The most widely used agent reasoning pattern. The LLM alternates between Thought (reasoning) and Action (tool call) steps until it reaches a Final Answer. Introduced by Yao et al. (2022).

Pattern: Thought → Action → Observation → Thought → Action → ... → Final Answer

✓ PROS

- ✓ Transparent — shows reasoning steps
- ✓ Works well with tool-heavy tasks
- ✓ Easy to debug (visible thought chain)
- ✓ Supported by all major frameworks
- ✓ Good performance on multi-step problems

✗ CONS

- ✗ Context grows with each step (token cost)
- ✗ Can get stuck in reasoning loops
- ✗ LLM may hallucinate tool inputs
- ✗ Slower than single-pass approaches
- ✗ Quality degrades over very long chains

■ *Best for: research tasks, data retrieval, code generation with verification, customer support with lookups*

2.2 Chain-of-Thought (CoT)

Prompts the LLM to reason step-by-step before producing an answer. Does not necessarily involve tool calls. Standard CoT: "Let's think step by step." Zero-shot CoT adds this prompt automatically. Few-shot CoT provides examples.

✓ PROS

- ✓ Dramatically improves accuracy on math/logic
- ✓ No external tools required
- ✓ Few-shot examples easy to craft
- ✓ Works with any LLM out of the box
- ✓ Reduces factual errors on complex reasoning

✗ CONS

- ✗ No grounding — relies on LLM knowledge only
- ✗ Still hallucinates on factual questions
- ✗ Longer prompts = higher cost
- ✗ No self-correction mechanism
- ✗ Quality highly model-dependent

■ *Best for: math, code reasoning, logical deduction, exam questions, structured analysis*

2.3 Tree of Thought (ToT)

Extends CoT by exploring multiple reasoning branches simultaneously. The LLM generates several candidate "thoughts" at each step, evaluates each, and prunes poor branches. Models a deliberate search process (BFS or DFS over thought space).

✓ PROS

- ✓ Best performance on complex planning problems
- ✓ Explores alternative approaches
- ✓ Self-evaluation built in
- ✓ Can backtrack from dead ends
- ✓ Finds solutions that linear CoT misses

✗ CONS

- ✗ Very high token consumption (5-10x CoT)
- ✗ Slow — multiple LLM calls per step
- ✗ Complex to implement correctly
- ✗ Expensive at scale
- ✗ Overkill for simple tasks

■ *Best for: creative writing, multi-step planning, game playing, complex code architecture, strategic decisions*

2.4 Self-Reflection / Self-Critique (Reflexion)

After producing an answer or taking an action, the agent reflects on its own output, identifies errors, and tries again. Reflexion (Shinn et al.) stores verbal feedback in episodic memory for future episodes. Also called "Self-RAG" when applied to retrieval.

✓ PROS

- ✓ Significantly improves accuracy over single-pass
- ✓ Catches its own mistakes
- ✓ Learns from failures within session
- ✓ Works without human feedback
- ✓ Combines well with ReAct and CoT

✗ CONS

- ✗ 2-3x more LLM calls (cost)
- ✗ May confidently "correct" correct answers
- ✗ Needs explicit stopping criteria
- ✗ Can loop endlessly without convergence check
- ✗ Reflection quality depends on model capability

■ *Best for: code generation, fact-checking pipelines, high-stakes decisions, quality-critical outputs*

2.5 Plan-and-Execute

Separates planning from execution. A planner LLM creates a full task plan upfront. An executor agent carries out each step. The plan can be revised dynamically if steps fail. Used in LangGraph, AutoGen, and other orchestration frameworks.

✓ PROS ✓ Clear structure — plan visible before execution ✓ Executor can be smaller/cheaper model ✓ Easy to parallelize independent steps ✓ Human can review/approve plan ✓ Good for long-horizon tasks	✗ CONS ✗ Initial plan may be wrong for dynamic environments ✗ Two-model coordination adds complexity ✗ Plan revision logic needs careful design ✗ Not ideal for exploratory/uncertain tasks ✗ Higher latency for first result
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■ *Best for: software development tasks, research workflows, data pipelines, multi-step automation*

2.6 LATS — LLM-Powered Monte Carlo Tree Search

Language Agent Tree Search combines ToT with Monte Carlo Tree Search (MCTS). The agent generates candidate actions, simulates outcomes, backtracks using a value function, and selects the best action. State-of-the-art on code generation benchmarks.

✓ PROS ✓ Highest quality on hard benchmarks ✓ Principled exploration-exploitation balance ✓ Finds globally optimal paths ✓ Value function guides search efficiently ✓ Backtracks intelligently from failures	✗ CONS ✗ Very high computational cost ✗ Complex to implement ✗ Slow — not for real-time use ✗ Requires good value/reward function design ✗ Niche — rarely needed outside research
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■ *Best for: competitive coding, complex mathematical proofs, strategic game playing, research*

3. AGENT ARCHITECTURES

How agents are structured and how they relate to each other fundamentally shapes capability, cost, maintainability, and failure modes.

3.1 Single Agent

One LLM with access to tools handles all tasks. The simplest architecture — a single ReAct or CoT loop with a tool registry. One LLM reasons, plans, picks tools, and produces output.

✓ PROS ✓ Simple to build and debug ✓ Low latency (no agent-to-agent communication) ✓ Easy to monitor — single thread ✓ Cost-predictable ✓ Sufficient for most practical tasks	✗ CONS ✗ Context window limits task complexity ✗ Cannot parallelize subtasks ✗ All tasks share one prompt/persona ✗ Hard to specialize behavior per domain ✗ Bottleneck on complex long-horizon tasks
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■ *Best for: simple Q&A; with tools, customer support bots, data lookup tasks, single-domain assistants*

3.2 Multi-Agent (Flat)

Multiple independent agents with different roles/tools collaborate. An orchestrator delegates tasks to specialized agents (researcher, coder, reviewer, writer). Agents communicate via shared message bus or direct messaging.

✓ PROS ✓ Specialization — each agent optimized for its role ✓ Parallelism — independent tasks run concurrently ✓ Scalable — add more agents for new capabilities ✓ Role-based system prompts improve quality ✓ Failures isolated to individual agents	✗ CONS ✗ Coordination overhead between agents ✗ Complex debugging across multiple agents ✗ Agent communication can introduce errors ✗ Higher token cost (multiple LLM calls) ✗ Risk of conflicting outputs requiring reconciliation
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■ *Best for: content creation pipelines, software development (coder+tester+reviewer), research synthesis*

3.3 Hierarchical (Manager + Worker)

A manager/supervisor agent breaks down goals and delegates to worker agents. Workers report back results. Manager synthesizes and decides next delegation. Can be multiple levels deep. Used in CrewAI and AutoGen hierarchical mode.

✓ PROS ✓ Clear responsibility separation ✓ Manager can adaptively re-plan based on worker output ✓ Workers can be specialized and lightweight ✓ Good for complex, multi-step goals ✓ Manager provides quality gate over workers	✗ CONS ✗ Manager is single point of failure ✗ Manager bottleneck limits parallelism ✗ Multi-level hierarchies add latency ✗ Manager needs strong capability (expensive model) ✗ Cascading failures if manager misunderstands
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■ *Best for: project management automation, code review pipelines, enterprise workflow automation*

3.4 Pipeline / Sequential

Agents arranged in a fixed sequence — output of Agent A becomes input of Agent B. Like Unix pipes but with LLM agents. Each agent performs one transformation. Deterministic flow, easy to reason about.

✓ PROS ✓ Predictable, easy to test each stage ✓ Each stage can use different model/prompt ✓ Simple to implement and monitor ✓ Each agent sees only relevant context (lower cost) ✓ Easy to replace individual stages	✗ CONS ✗ No parallelism — strictly sequential ✗ Errors propagate downstream ✗ Rigid — cannot adapt flow based on results ✗ Not suitable for exploratory tasks ✗ Latency = sum of all stage latencies
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■ *Best for: document processing, ETL pipelines, content moderation, translation + summarization chains*

3.5 Swarm (Decentralized)

Large number of lightweight agents operate without a central coordinator. Agents communicate peer-to-peer, share a common memory/state, and self-organize. Inspired by ant colonies and particle swarm optimization. OpenAI's Swarm framework.

✓ PROS ✓ Highly scalable — add/remove agents dynamically ✓ No single point of failure ✓ Emergent behavior from simple rules ✓ Naturally parallel ✓ Resilient to individual agent failures	✗ CONS ✗ Emergent behavior hard to predict/control ✗ Debugging extremely difficult ✗ No guarantee of convergence ✗ Coordination via shared state creates race conditions ✗ Not suitable for tasks needing coherent reasoning
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■ *Best for: simulation, distributed data collection, adversarial testing, exploration tasks*

3.6 Graph-Based (State Machine)

Agents and their transitions modeled as a directed graph (state machine or DAG). LangGraph is the primary framework. Each node is an agent or function. Edges define conditional routing. Supports cycles (loops) and branching.

✓ PROS ✓ Precise control over agent flow ✓ Conditional branching and looping supported ✓ Visual representation of workflow ✓ Easy to add human-in-the-loop checkpoints ✓ Testable — each node independently	✗ CONS ✗ More code to define graph structure ✗ Can become complex for large workflows ✗ Requires upfront workflow design ✗ Less flexible than purely dynamic approaches ✗ Learning curve for LangGraph API
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■ *Best for: complex business workflows, approval chains, multi-step automation with conditions*

4. MEMORY SYSTEMS

Memory determines how much context an agent can use and how it persists knowledge across interactions. The right memory architecture is critical for agent performance.

Memory Type	Storage	Scope	Best For
In-Context (Working)	LLM context window	Current session	Short tasks, conversation history
External Vector DB	Pinecone, Weaviate, ChromaDB	Persistent	Long-term knowledge, RAG
Episodic Memory	DB of past episodes	Persistent	Learning from past interactions
Semantic Memory	Knowledge graph / Vector DB	Persistent	Facts, entities, relationships
Procedural Memory	Tool/skill registry	Persistent	How-to knowledge, tool usage
Cache / KV Store	Redis, Memcached	Session/TTL	LLM prompt caching, result caching

4.1 In-Context Memory

Everything in the LLM's context window. The simplest approach — just append history, tool results, and observations to the prompt. Limited by context window size (4K–1M tokens).

✓ PROS

- ✓ Zero infrastructure needed
- ✓ Perfect recall of everything in window
- ✓ No retrieval latency
- ✓ No chunking/embedding errors

✗ CONS

- ✗ Hard limit on context size
- ✗ Cost grows linearly with context length
- ✗ Attention dilution on very long contexts
- ✗ Lost on session end

4.2 Vector Database (External Memory)

Encode memories as embedding vectors. Store in vector DB. At query time, retrieve most relevant memories via semantic similarity search. Foundation of RAG-based agents. Databases: Pinecone, Weaviate, ChromaDB, Qdrant, pgvector (PostgreSQL).

✓ PROS

- ✓ Unlimited storage capacity
- ✓ Semantic search (not just keyword matching)
- ✓ Persistent across sessions
- ✓ Scale independently of context window
- ✓ Enables true long-term memory

✗ CONS

- ✗ Retrieval quality depends on embedding model
- ✗ Chunking strategy affects recall
- ✗ Infrastructure overhead
- ✗ Retrieval latency (10-100ms)
- ✗ May miss relevant context (top-k retrieval is imperfect)

5. TOOL USE & FUNCTION CALLING

Tools are what separate agents from chatbots. The ability to take actions in the real world — search the web, run code, query databases, call APIs — is the core superpower of agentic systems.

5.1 Tool Design Principles

- Single responsibility: each tool does one thing well
- Clear descriptions: LLM selects tools based on description — make it unambiguous
- Typed parameters: use JSON Schema for parameter validation
- Atomic operations: prefer tools that are safe to retry (idempotent)
- Return structured data: JSON responses easier for LLM to parse than free text
- Error messages must be informative: LLM needs to understand failure to retry correctly

5.2 Common Tool Categories

Category	Examples	Key Considerations
Search / Web	Tavily, Serper, Bing API, Wikipedia	Rate limits, result quality, cost per query
Code Execution	Python REPL, E2B sandbox, Jupyter	Sandboxing critical — never run untrusted code on host
Database	SQL queries, NoSQL, vector search	Read-only vs read-write permissions, injection risks
File System	Read/write files, parse PDFs, CSVs	Path traversal attacks, size limits
APIs / Services	REST APIs, email, calendar, Slack	Auth handling, rate limits, side effects
Browser / UI	Playwright, Selenium, Puppeteer	Headless browser cost, fragility of UI selectors
LLM Sub-calls	Summarizer, translator, classifier	Nested cost, prompt injection between agents
Memory Tools	Save/retrieve from vector DB	Embedding quality, retrieval relevance tuning

5.3 Function Calling vs. ReAct Tool Use

Aspect	OpenAI Function Calling	ReAct Text-Based Tool Use
Mechanism	Native API support — JSON structured output	LLM outputs action in text, parsed by framework
Reliability	Higher — structured, validated	Lower — parsing can fail
Model support	GPT-4, Claude, Gemini, Mistral	Any instruction-following LLM
Parallel calls	Yes — multiple tools in one response	Sequential by default
Flexibility	Constrained to defined schema	More flexible, less structured

6. ORCHESTRATION FRAMEWORKS

Frameworks abstract the boilerplate of agent loops, tool management, memory, and multi-agent coordination. Choosing the right one significantly impacts developer velocity and production reliability.

6.1 LangChain

The most widely adopted agent framework. Modular chains of LLMs, prompts, tools, memory, and retrievers. Huge ecosystem of integrations (200+ tools). AgentExecutor for ReAct agents. LCEL (LangChain Expression Language) for composition.

✓ PROS ✓ Massive ecosystem and community ✓ Extensive integrations (tools, DBs, LLMs) ✓ Good documentation and examples ✓ Supports all major LLMs ✓ Active development and regular releases	✗ CONS ✗ Abstraction can hide what's happening ✗ Can be over-engineered for simple tasks ✗ Version breaking changes frequent ✗ Debug complexity in chained components ✗ Performance overhead from abstraction layers
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■ *Best for: RAG applications, tool-using agents, teams wanting rich ecosystem*

6.2 LangGraph

Built on LangChain — models agent workflows as stateful graphs (nodes + edges). Each node is a function or agent. Edges define conditional routing. Supports cycles (loops), human-in-the-loop, and streaming. Best for complex multi-agent workflows.

✓ PROS ✓ Fine-grained control over agent flow ✓ First-class support for human-in-the-loop ✓ Streaming built in ✓ Checkpointing for long workflows ✓ Parallel node execution ✓ Visual graph representation	✗ CONS ✗ More verbose than simple chains ✗ Steeper learning curve than basic LangChain ✗ Still relatively new — APIs evolving ✗ Requires graph thinking upfront ✗ Debugging state across nodes requires tooling
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■ *Best for: complex multi-step workflows, approval chains, stateful long-running agents*

6.3 AutoGen (Microsoft)

Multi-agent conversation framework from Microsoft Research. Agents communicate via natural language messages in a group chat. Supports: AssistantAgent (LLM), UserProxyAgent (human/code executor), GroupChatManager (orchestrator). Built-in code execution with Docker sandboxing.

✓ PROS ✓ Natural multi-agent conversations ✓ Built-in code execution + Docker sandbox ✓ Human proxy agent for easy human-in-loop ✓ Strong for coding and research tasks ✓ Microsoft Research backing, active development	✗ CONS ✗ Conversation-based — harder to control precise flow ✗ Can be verbose (lots of agent chat) ✗ Less ecosystem than LangChain ✗ Token-heavy for long multi-agent chats ✗ GroupChat coordination can be unpredictable
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■ *Best for: coding assistants, research automation, tasks requiring code execution*

6.4 CrewAI

Role-based multi-agent framework. Define Agents (role, goal, backstory), Tasks (description, expected output, agent), and Crew (process: sequential or hierarchical). Built on LangChain. Simple, opinionated API. Great for quick multi-agent prototyping.

✓ PROS ✓ Very simple, intuitive API ✓ Role-based agents feel natural ✓ Sequential and hierarchical processes out of box ✓ Good for non-expert users ✓ Fast to prototype multi-agent systems	✗ CONS ✗ Less control than LangGraph ✗ Opinionated — harder to customize deeply ✗ Built on LangChain (inherits its complexity) ✗ Less suitable for complex conditional flows ✗ Memory management less mature
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■ *Best for: content creation teams, research pipelines, business process automation prototypes*

6.5 Semantic Kernel (Microsoft)

Enterprise-focused SDK for integrating LLMs into existing applications. Available for Python, C#, Java. Concepts: Kernel (central orchestrator), Plugins (tool collections), Planner (automatic task planning), Memory (vector stores). Strong .NET/Java integration — popular in enterprise Microsoft shops.

✓ PROS ✓ Multi-language: Python, C#, Java ✓ Strong enterprise and Azure integration ✓ Planner generates plans from goals ✓ Plugin system is clean and extensible ✓ Good documentation for enterprise patterns	✗ CONS ✗ Less community content than LangChain ✗ C# version more mature than Python ✗ More complex than LangChain for simple tasks ✗ Azure-centric (less cloud-agnostic) ✗ Planner reliability can be inconsistent
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■ *Best for: Java/.NET enterprise apps, Azure-native deployments, large org AI platforms*

6.6 LlamaIndex (Agentic RAG)

Originally a RAG framework, now also supports agentic patterns. Best-in-class for data ingestion, indexing, and retrieval. AgentRunner + ReActAgent. Strong abstractions for documents, structured data, and multi-modal data.

✓ PROS ✓ Best-in-class RAG capabilities ✓ Excellent data connectors (100+) ✓ Query engines for structured + unstructured data ✓ Multi-modal support ✓ SubQuestion and RouterQueryEngine for complex RAG	✗ CONS ✗ Agent capabilities less mature than LangChain ✗ Primarily RAG-focused ✗ Smaller community than LangChain ✗ More complex for non-RAG use cases ✗ Documentation can be inconsistent
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■ *Best for: document-heavy agents, knowledge base Q&A, enterprise data retrieval*

6.7 Framework Comparison Summary

Framework	Ease of Use	Control	Multi-Agent	RAG	Enterprise	Best For
LangChain	Medium	Medium	Via LangGraph	★★★★★	★★★	General purpose
LangGraph	Complex	High	★★★★★	Via LangChain	★★★★	Complex workflows
AutoGen	Medium	Medium	★★★★★	★★	★★★	Code + research
CrewAI	Easy	Low	★★★★	★★★	★★	Quick prototypes
Semantic Kernel	Medium	High	★★★	★★★	★★★★★	Enterprise/.NET/Java
LlamaIndex	Medium	Medium	★★	★★★★★	★★★	Document RAG

7. RAG (RETRIEVAL-AUGMENTED GENERATION) FOR AGENTS

RAG grounds agents in external knowledge, dramatically reducing hallucination and enabling use of private/up-to-date data. The evolution from basic RAG to Agentic RAG represents a major capability leap.

7.1 Naive RAG

Simple retrieve-then-generate: embed query, find top-k similar chunks in vector DB, stuff into prompt, generate answer. Fast to implement. Widely used baseline.

✓ PROS ✓ Simple to implement ✓ Fast (single retrieval step) ✓ Works well for simple Q&A ✓ Low latency ✓ Easy to debug	✗ CONS ✗ Single retrieval step may miss relevant context ✗ Chunk size/overlap tuning required ✗ No query reformulation ✗ Multi-hop questions fail ✗ No validation of retrieved content quality
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7.2 Advanced RAG

Adds pre-retrieval and post-retrieval enhancements: query rewriting, HyDE (Hypothetical Document Embeddings), re-ranking (Cohere Rerank, CrossEncoder), multi-query retrieval, contextual compression, and parent document retrieval.

✓ PROS ✓ Significantly better retrieval quality ✓ Handles ambiguous queries better ✓ Re-ranking reduces noise in top-k results ✓ Parent chunking improves context ✓ Multi-query covers query variations	✗ CONS ✗ More latency (extra retrieval/rerank steps) ✗ Higher cost (extra LLM/reranker calls) ✗ More components = more failure points ✗ Tuning more complex ✗ Reranker adds infrastructure dependency
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7.3 Agentic RAG

The agent dynamically decides WHEN and HOW to retrieve. It can: issue multiple retrieval queries, use different indexes, follow up on retrieved results, decide if retrieved info is sufficient, and combine retrieved context with web search. Transforms RAG from a single step into a reasoning process.

✓ PROS ✓ Adaptive retrieval — retrieves only what's needed ✓ Handles multi-hop questions naturally ✓ Can combine multiple knowledge sources ✓ Self-corrects poor retrieval ✓ Much better on complex research tasks	✗ CONS ✗ Multiple LLM calls = higher latency and cost ✗ Complex to implement reliably ✗ Harder to predict behavior ✗ Risk of over-retrieval loops ✗ Requires strong reasoning model
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■ *The gold standard for knowledge-intensive agents. Use LangGraph or LlamaIndex for implementation.*

7.4 GraphRAG (Microsoft)

Builds a knowledge graph from documents. Retrieval traverses graph relationships rather than just vector similarity. Excels at questions requiring relationship understanding across document corpus. Can answer "how does X relate to Y across all documents?"

✓ PROS ✓ Captures relationships between entities ✓ Better for global/thematic questions ✓ Excels at cross-document synthesis ✓ Richer context than flat chunks ✓ Good for corporate knowledge bases	✗ CONS ✗ Expensive to build the graph upfront ✗ Complex infrastructure (graph DB) ✗ Slow indexing for large corpora ✗ Higher query latency ✗ Overkill for simple document Q&A
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8. SAFETY & RELIABILITY

Agents that take real-world actions require careful safety engineering. The consequences of agent errors range from bad answers to deleted data, sent emails, or financial transactions.

8.1 Key Safety Risks

Risk	Description	Mitigation
Prompt Injection	Malicious content in tool output hijacks agent	Sanitize tool output, use separate context for untrusted data
Hallucination	Agent confidently acts on false information	RAG grounding, self-reflection, verification tools
Infinite Loops	Agent loops without reaching goal	Max iteration limit, loop detection, timeout
Unintended Side Effects	Agent takes destructive real-world actions	Read-only tools first, human approval for write actions
Scope Creep	Agent does more than instructed	Clear system prompt scope, tool permission limits
Data Leakage	Agent exposes sensitive data	PII detection, output filtering, data access controls
Cost Explosion	Runaway agent burns token budget	Token budgets, cost tracking, circuit breakers

8.2 Guardrail Strategies

- Input guardrails: validate and sanitize user input before it reaches the agent
- Output guardrails: check agent output for harmful content, PII, or policy violations
- Tool permissions: principle of least privilege — agents get only tools they need
- Confirmation gates: require human approval before irreversible actions (send email, delete record)
- Max iterations: hard limit on agent loop cycles (e.g., 15 max steps)
- Budget controls: token and cost limits per agent run
- Sandboxing: execute code in isolated containers (E2B, Docker) not on host
- Logging: log every thought, action, and observation for audit trail

8.3 Testing AI Agents

- Unit test each tool independently with mock inputs
- Integration test the full agent loop with controlled scenarios
- Adversarial testing: try to break the agent with edge cases and injection attempts
- Regression traces: save agent traces, compare behavior across model/prompt updates
- Evaluation datasets: build gold-standard test cases with expected tool calls + outputs
- LLM-as-judge: use a separate LLM to evaluate agent output quality
- Load testing: measure agent performance under concurrent requests

9. PRODUCTION DEPLOYMENT

Building an agent that works in a demo is easy. Building one that works reliably at scale in production requires engineering discipline around observability, cost, latency, and failure handling.

9.1 Observability

- Trace every agent run: record all LLM calls, tool calls, inputs, outputs, timing
- Tools: LangSmith (LangChain), Langfuse, Helicone, Arize Phoenix, OpenTelemetry
- Metrics: agent success rate, average steps per run, tool failure rate, latency p50/p99, cost per run
- Alerts: failed runs, cost spikes, latency degradation, unexpected tool call patterns
- Human review queue: sample N% of runs for quality review

9.2 Cost Management

- Token budgets: set max input+output tokens per agent run
- Model tiering: use GPT-4o-mini / Haiku for simple subtasks, reserve GPT-4o / Sonnet for complex reasoning
- Caching: cache LLM responses for identical prompts (semantic caching with GPTCache)
- Prompt optimization: shorter system prompts, compressed context, message pruning
- Batch processing: aggregate non-urgent agent tasks for batch API (50% cost reduction)
- Cost per run budget: circuit break if single run exceeds \$N threshold

9.3 Latency Optimization

- Streaming: stream tokens back to user while agent continues processing
- Parallel tool calls: execute independent tool calls simultaneously
- Smaller models for classification/routing steps (sub-100ms)
- Tool result caching: cache expensive tool calls with TTL
- Async architecture: non-blocking agent execution, webhook/polling for long runs
- Target: p50 < 3s for simple tasks, p50 < 30s for complex multi-step tasks

9.4 Failure Handling

- Retry with exponential backoff on LLM API rate limits and transient errors
- Tool-level retries: retry failed tool calls with reformulated inputs
- Fallback models: switch to backup LLM on primary model failure
- Graceful degradation: return partial result if agent fails to complete fully
- Dead letter queue: persist failed agent runs for manual review and retry
- Idempotency keys: ensure agent runs can be safely retried without side effects

10. MODEL SELECTION FOR AGENTS

Model capability is the biggest determinant of agent quality. The right model depends on task complexity, latency requirements, cost constraints, and data privacy.

Model	Provider	Strengths	Best Agent Use Case
GPT-4o	OpenAI	All-round, strong tool use, vision	General-purpose production agents
GPT-4o mini	OpenAI	Fast, cheap, good tool use	High-volume, cost-sensitive agents
Claude Sonnet 4	Anthropic	Long context, careful reasoning, coding	Research, coding, long-document agents
Claude Haiku	Anthropic	Ultra-fast, cheap	Classification, routing, simple tasks
Gemini 1.5 Pro	Google	1M token context, multimodal	Very long context, video/image tasks
Llama 3.1 70B	Meta (Open)	Open source, self-hostable	On-prem, privacy-sensitive deployments
Mistral Large	Mistral	European data residency, efficient	EU compliance, cost-efficient agents
DeepSeek R1	DeepSeek	Strong reasoning, very low cost	Math, code, structured reasoning

11. REAL-WORLD USE CASES

Use Case	Strategy	Framework	What It Does
Software Development Agent	ReAct + Code Exec + Self-Reflection	LangGraph + AutoGen	Reads requirements, writes code, runs tests, fixes failures, commits PR
Research / Analysis Agent	Plan-and-Execute + ReAct	LangChain + LlamaIndex	Searches web, reads papers, synthesizes findings, writes report
Customer Support Agent	ReAct + RAG	LangGraph + Zendesk tools	Understands query, retrieves knowledge base, handles escalation
Data Analysis Agent	ReAct + Code Exec	LangChain + Pandas + SQL tools	Understands question, queries DB, runs analysis code, produces charts
Document Processing Agent	Pipeline + RAG	LlamaIndex + LangGraph	Extracts, classifies, summarizes, routes documents automatically
Financial Analysis Agent	ReAct + Self-Reflection	Semantic Kernel + LoanIQ SDK	Queries loan data, calculates metrics, generates risk reports
DevOps / SRE Agent	ReAct + Plan-Execute	LangGraph + k8s/cloud tools	Monitors alerts, diagnoses issues, runs remediation playbooks
HR / Recruitment Agent	Pipeline + Multi-Agent	CrewAI + ATS integration	Screens resumes, schedules interviews, generates feedback

12. STRATEGY SELECTION GUIDE

Use this decision guide to select the right combination of reasoning strategy, architecture, and framework for your use case.

Decision Tree

- Simple Q&A with factual lookup? → Single Agent + ReAct + RAG (LlamaIndex/LangChain)
- Multi-step task with multiple tools? → Single Agent + ReAct or Plan-Execute (LangChain/LangGraph)
- Complex workflow with conditions/approvals? → Graph-based (LangGraph) + Human-in-loop
- Different roles needed (researcher, writer, reviewer)? → Multi-Agent (CrewAI or AutoGen)
- Enterprise Java/.NET app? → Semantic Kernel
- Code generation with execution? → AutoGen or LangGraph + code sandbox
- Document-heavy knowledge base? → LlamaIndex + Advanced RAG + Agentic RAG
- Privacy/on-prem required? → Open source model (Llama 3) + self-hosted framework
- High accuracy on complex reasoning? → ToT or LATS + self-reflection
- Speed and cost priority? → Single Agent + GPT-4o-mini/Haiku + prompt caching

Cost vs. Quality Trade-offs

Scenario	Recommended Approach	Estimated Cost
Prototype / POC	CrewAI or LangChain + GPT-4o-mini	Very Low
High-volume production	Single Agent + GPT-4o-mini + caching	Low
Quality-critical tasks	Plan-Execute + GPT-4o + self-reflection	Medium
Complex multi-agent	LangGraph + mixed models (cheap for simple, powerful for complex)	Medium-High
Research / max quality	ToT/LATS + GPT-4o + multiple reflection rounds	High

13. BEST PRACTICES & LESSONS LEARNED

System Prompt Engineering

- Define agent role, personality, and constraints clearly in system prompt
- Specify output format explicitly (JSON, markdown, plain text)
- Include examples of good tool usage in the system prompt
- Add explicit instructions for handling uncertainty ("say I don't know rather than guess")
- State stopping conditions — when should the agent consider task complete?

Tool Design

- Write tool descriptions as if explaining to a human — the LLM reads them verbatim
- Start with read-only tools; add write tools only when necessary
- Return errors in a format the LLM can act on ("No results found for query X")
- Limit tool count per agent — too many tools confuse the model (<10 is ideal)
- Group related tools into one tool with sub-actions rather than many tiny tools

Reliability

- Always set max_iterations (10-20 typical) to prevent infinite loops
- Implement circuit breakers on expensive external tool calls
- Store agent state for recovery — agents can be long-running
- Design for idempotency — agent runs should be safely retryable
- Test with adversarial inputs — what happens if a tool returns garbage?

Human-in-the-Loop

- Add approval gates before any irreversible action (delete, send, pay)
- Provide clear summaries of what the agent plans to do before execution
- Allow human to edit agent plan, not just approve/reject
- Escalation path: if agent stuck after N retries, escalate to human
- Audit trail: every action should be attributable and reversible where possible

Evaluation

- Build a golden dataset of test cases with expected outcomes before shipping
- Measure: task success rate, average steps, average cost, latency distribution
- Use LLM-as-judge to evaluate subjective quality (faithfulness, helpfulness, safety)
- Regression test on every prompt/model change
- A/B test new agent versions on small % of traffic before full rollout

14. FUTURE TRENDS IN AI AGENTS

- Longer context windows (Gemini 1.5: 1M tokens) reducing need for external memory
 - Native multimodal agents — see, hear, act across text, images, video, audio
 - Specialised agent models (fine-tuned for tool use, planning, code execution)
 - Agent-to-Agent protocols — standardized communication between agents from different vendors
 - Persistent agents — agents that run indefinitely, learn from every interaction
 - Agent marketplaces — plug-and-play specialized agents as microservices
 - Physical world agents — robotics + LLM planning (Boston Dynamics + GPT-4V)
 - Regulatory frameworks — EU AI Act and equivalent will impose agent accountability rules
 - Cost commoditization — inference costs dropping 10x/year, enabling more agent calls
 - Self-improving agents — agents that write and deploy their own tool improvements
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This guide covers the state of AI Agent development as of 2025-2026. The field evolves extremely fast — follow LangChain blog, Anthropic research, OpenAI docs, and arXiv (cs.AI) for the latest advances.

Key arXiv papers to read: ReAct (2210.03629), Reflexion (2303.11366), ToT (2305.10601), LATS (2310.04406), Self-RAG (2310.11511), GraphRAG (Microsoft 2024)